CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

May 2014

CL209 Structural Analysis-I

Date: 05.05.2014, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What is idealization of a structure? Explain with suitable examples. **[03]**

(b) Enlist four advantages of statically determinate structures. **[04]**

Q - 2 (a) A cylindrical shell 3m long and 1m internal diameter is subjected to an internal pressure of 1 N/mm^2. If the thickness of the shell is 12mm, find the circumferential & longitudinal stresses. Also find the maximum shear stress & change is dimensions of the shell. **[07]**

(b) Attempt any One **[07]**

(i) A column is made up of a T Section. Its flange is 200*20mm where as web is 180*15mm. A load of 100kN acts at a point on the centroidal axis, 40mm below the centroidal x-x axis. Calculate the maximum and minimum stresses induced in the section.

(ii) Explain limit of eccentricity and no tension condition with suitable examples.

Q - 3 Attempt any Two **[14]**

(i) A beam of 6m length, simply supported at ends A and B is loaded with two point loads of 60kN and 50 kN at distance 1m and 3m from left support A. Determine the position and magnitude of maximum deflection. Take $E = 2*10^5$ N/mm^2 and $I = 8500cm^4$. (Use Macaulay's Method)

(ii) Beam ABCD is having length of L m. Span BC is simply supported and AB and CD are overhanging portions. Span BC is of length l m. Find ratio of L/l so that the upward deflection at each end equals the down word deflection at mid span, due to a central point load.

(iii) For a simply supported beam ACDB, AC = L/4 & I = I, CD = L/4 & I = 2I and DB = L/2 & I = I. Find deflection at the midpoint if beam is loaded with point load of P at point C. Use moment area method.

SECTION – II

Q - 4 Compare crippling load of a solid circular section of 250mm diameter with a hollow circular section of same sectional area and 40mm thickness. The other conditions are same for both the columns **[07]**

Q - 5 (a) Derive Euler's formula for crippling load of a column having both ends hinged. **[04]**

(b) Attempt any One **[10]**

(i) A three hinge parabolic arch hinged at the supports and at the crown has a span of 24m and a central rise of 4m. It carries a concentrated load of 50kN at 18m from left support and a uniformly distributed load of 30kN/m over the left half portion. Determine the moment, thrust and radial shear at a section 6m from the left support.

(ii) Show that the parabolic shape is a funicular shape for a three-hinge arch subjected to a uniformly distributed load over to its entire span.

Q - 6 Attempt any Two **[14]**

(i) Four point loads, 8, 15, 15 and 10 kN have c/c spacing of 2m between consecutive loads and they traverse a girder of 30m span from left to right with 10kN load as leading load. Calculate the maximum bending moment and positive shear force at 8m from the left support.

(ii) An udl of 12 kN/m and 3m length crosses a simply supported girder of span 10m from left to right. Draw influence line for S.F. and B.M. at section 4m from left and find maximum positive S.F. and B.M. at this section.

(iii) Four equal loads of 80 kN each, equally spaced at 2m apart followed by a uniformly distributed load of 60 kN/m run at a distance 2m from the last 80 kN load cross a girder of 20m span from right to left. Using influence lines, calculate maximum S.F. and B.M. at a section 8m from the left hand support when the leading 80 kN load is 5m from this support.

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